

Item	Definition	Notes	Additional note
Study type	Lifecycle scenario comparison	This is not a complete LCA	
Case	VELFAC 200 ENERGY - Top-guided residential window	Publicly documented manufacturer case	
Functional unit	Providing one residential window opening of defined dimensions and equivalent thermal, acoustic, safety and weather-protection performance over 50 years	Based on the project protocol	
Reference dimensions	1.23 x 1.48 m	Based on the VELFAC EPD reference window	
Study period	50 years	Fixed project boundary	
Building type	Residential	Fixed project boundary	
Geographic context	Europe	Consistent with project scope and selected case	
Compared scenarios	Scenario A: full replacement	Scenario B: component repair and replacement	Scenario C is excluded because there is not enough evidence for a robust reuse or refurbishment scenario
Main purpose	To compare whether full replacement or component-level repair and replacement is more plausible as a product-life extension strategy	The comparison is qualitative and assumption-based	
Claim level	Exploratory	Not validated; not a quantified environmental-impact assessment	
Main limitation	No complete product-specific use-phase, repair-frequency, transport or end-of-life data are available	Therefore the comparison identifies plausible trade-offs and uncertainties rather than exact environmental savings	